

CRT - Answers are bolded

1. What is the purpose of a loop structure?

It repeats a process over and over again. Something like, summing of numbers or prompting the user over and over again until a valid outcome is reached.

2. Explain the difference between a while statement and a do-while statement.

A do-while statement does not evaluate the condition until AFTER the first execution of the loop

3. An input validation loop is a loop that checks user input for valid data. If valid data is not entered, the loop iterates until valid data is entered. In which review of this chapter did you write code for an input validation loop?

As per D2I i dont believe we wrote an input validation loop code.

4.

a) What is an infinite loop?

A loop that continues forever

b) List two types of errors that can lead to an infinite loop.

Syntax and logic errors

c) What is meant by overflow?

When there are not enough bits to store a number.

5. How many times will the do - while loop execute?

```
int x = 0;  
do {  
    x = x + 2;  
} while (x < 120);
```

I believe that the loop will go around 60 times.

6. What initial value of x would make the loop

infinite?

```
do {  
    x = x - 3;  
} while (x < 120);
```

Any number above less than 123 will probably make the loop infinite and probably hang as mentioned in the textbook.

7. Compare and contrast counters and accumulators. List two uses for each.

Counters do exactly what's in their name, counting the number of times something has happened while accumulators STORE a total etc.

Counters uses : Displaying pages, creating lists like the outcomes from english (1.1,1.2,1.3)etc

Accumulator uses: Tracking the weight of something, tracking the game score.

8. Write a for statement that sums the integers from 3 to 10, inclusive.

```
int maxValue = 10;  
int added = 0;  
  
    for (int newValue = 3; newValue <= maxValue;  
  
newValue++) {  
        added += newValue;  
    }  
System.out.println("The sum of the integers is : " + added);
```

9. List two factors that should be considered when determining which loop structure to choose.

The 2 factors that should be considered are Check if the loop should run at least 1 time before writing the code and estimating how many times it should run

11. Consider the following assignment: String x = "my string."; Determine the value returned by each of the following methods:

a) x.length()

Gives a 10

b) `x.substring(0, 3)`

Will give my (my and a space

c) `x.toLowerCase()`

my string, because its already lower case

d) `x.toUpperCase`

MY STRING, it just makes it uppercase

e) `x.trim()`

**This removes whitespace, and we don't have any so same output
my string**